PAPER_784: M82 Cigar Galaxy — UQFF Starburst Superwind

Author: Daniel T. Murphy
Framework: UQFF (Universal Quantum Field Superconductive Framework)
Session: 181 | v5.42
Date: 2026
CP4 Class: #368 — M82CigarStarburstUQFFCalculator

Abstract

M82 (NGC 3034), the "Cigar Galaxy," is the archetypal starburst galaxy, located only ~12 million light-years away ($z \approx 0.0008$) in Ursa Major. Tidally disturbed by its companion M81, M82 experiences a star-formation rate roughly 10× higher than the Milky Way, driving a spectacular bi-polar superwind of hot gas and dust erupting ~12 kly above and below the disk. The superwind magnetic field reaches $\sim 10^{-4}$ T — characteristic of the starburst regime. Under UQFF, $v = 10^6$ m/s (superwind velocity) and $B = 10^{-4}$ T (starburst-amplified field) yield g_M82 $\approx 1.053\times10^{-1}$ m/s^2, identical to the Tarantula Nebula and Stephan's Quintet at these extreme parameters.

---

1. Introduction

M82's starburst was triggered ~100 Myr ago by a close encounter with M81. The resulting disk starburst currently produces ~10 M⊙/yr in a region only ~1 kpc in diameter — one of the most concentrated starbursts in the nearby universe. The galactic-scale superwind reaches ~1,000 km/s and carries a luminosity of $\sim 10^{41}$ erg/s. Radio measurements confirm B-fields of ~50–200 μT throughout the starburst disk. UQFF encodes the superwind through $v = 10^6$ m/s and the starburst-amplified $B = 10^{-4}$ T, placing M82 in the UQFF starburst regime alongside Tarantula 30 Dor (PAPER_774) and Stephan's Quintet (PAPER_778).

---

2. Master UQFF Gravity Equation

```
g_M82(r, t) = (G × M) / r² × (1 + H(z)×t) × (1 + M_sf) × (1 + f_TRZ) + a_EM
```

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	10^{10} M⊙ = 1.989×10^{40} kg	NED
Disk radius	r	2×10^{20} m (~21 kly)	NED
SFR	—	10 M⊙/yr	Radio/IR
Age	t	1×10^{8} yr = 3.156×10^{15} s	Starburst duration
M_sf	—	0.15	UQFF starburst mass fraction
Redshift	z	0.0008	Spectroscopic
v_EM	v	10^6 m/s	Superwind velocity
B_EM	B	10^{-4} T	Starburst-amplified B
f_TRZ	—	0.05	UQFF

---

3. Long-Form Derivation

Step 1: Base Gravitational Term

$g\_grav = 6.6743e\text{-}11 \times 1.989e40 / (2e20)^2 = 3.319e\text{-}11\ m/s^2$

Step 2: Cosmic Expansion Factor

$H(z) = 2.268e\text{-}18\ s^{-1}$; $H(z)\times t = 2.268e\text{-}18 \times 3.156e15 = 7.160e\text{-}3$; factor = 1.00716

Step 3: SFR Mass Fraction (Starburst)

M_sf = 0.15; 1 + M_sf = 1.15

Step 4: Time-Reversal Correction

f_TRZ = 0.05; 1 + f_TRZ = 1.05

Step 5: Gravitational Total

$g\_grav\_total = 3.319e\text{-}11 \times 1.00716 \times 1.15 \times 1.05 = 4.015e\text{-}11\ m/s^2$

Step 6: Aether EM Correction (Starburst Level)

$v = 10^6$ m/s, $B = 10^{-4}$ T
$a\_EM = (1.602e\text{-}19 \times 10^6 \times 10^{-4} / 1.673e\text{-}27) \times 11 \times 10^{-12} = 1.053e\text{-}1\ m/s^2$

Step 7: Final Solution

$g\_M82 = 4.015e\text{-}11 + 1.053e\text{-}1 \approx 1.053e\text{-}1\ m/s^2$

---

4. Physical Interpretation

At only 12 Mly distance, M82 is the closest prototype of the starburst superwind regime. The observed superwind velocity ~1,000 km/s and starburst B-field ~10^{-4} T both directly confirm the UQFF starburst parameters. M82's result $g = 1.053\times10^{-1}$ m/s^2 matches Tarantula Nebula (PAPER_774) and Stephan's Quintet (PAPER_778), confirming UQFF universality across: dwarf-scale starburst (30 Dor in LMC), compact-group intergalactic shock (Stephan's Quintet), and galaxy-scale starburst (M82) at the same extreme EM parameter combination ($v = 10^6$ m/s, $B = 10^{-4}$ T).

---

5. Conclusions

UQFF applied to M82 yields $g \approx 1.053\times10^{-1}$ m/s^2, confirming M82 occupies the same UQFF starburst-shock class as Tarantula and Stephan's Quintet. At z = 0.0008, the nearest starburst galaxy serves as the closest-distance validation point for the UQFF starburst regime.

PAPER_784, CP4 class #368. v5.42.